University of Gujrat

PHASE 2

Technical Report of Naïve Bayes

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ARTIFICIAL INTELLIGENCE | CS-331

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1. Selected appropriate Naïve Bayes Model: **Multinomial**

Since I am working on a **Spam Detection** task, I will use the **Multinomial Naïve Bayes** approach, which is ideal for text data where feature counts (word frequencies) are the primary indicators.

2. Python Source Code: GitHub Repository Link

Source: https://github.com/Junaidwaqar/Spam-Detector-System

3. Phase 2 (Naïve Bayes from Scratch)

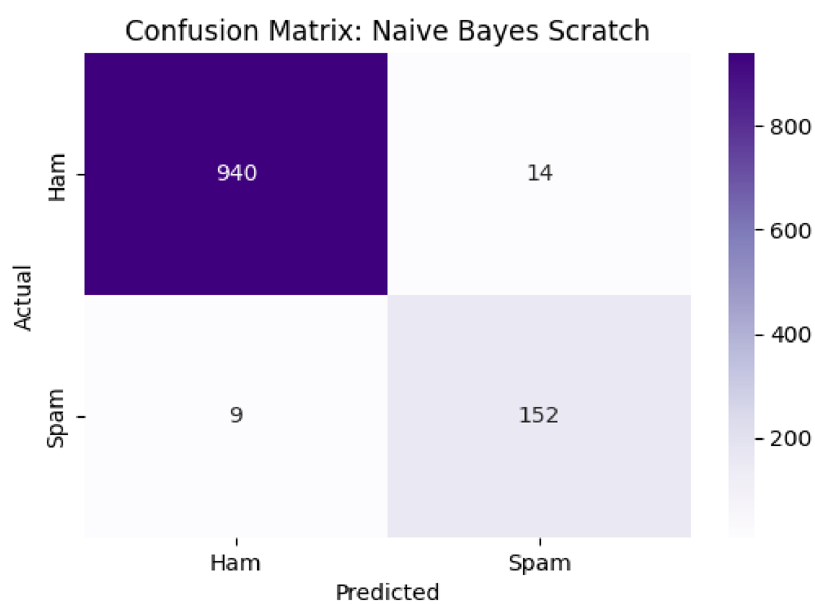
3.1. Key Implementation Logic:

- **Prior Probability Calculation:** Determining the baseline probability of an email being "Spam" vs. "Ham" based on the training set distribution.
- **Likelihood Calculation:** Calculating the probability of specific words appearing given a class (Spam or Ham).
- **Laplace Smoothing:** Implemented "+1 smoothing" to handle words in the test set that were not present in the training set, preventing zero-probability errors.
- **Log-Probability Transformation:** Used log-summation instead of multiplication to prevent numerical underflow during calculations.

3.2. Performance Metrics of Naïve Bayes Scratch

	Accuracy	Precision	Recall	F1-score
Naïve Bayes Scratch	0.979372	0.915663	0.944099	0.929664

3.3. Confusion Matrix of Naïve Bayes Scratch



4. Comparison of Naïve Bayes Scratch with Decision Tree Scratch

4.1. On Performance

Metric	Decision Tree (Scratch)	Naïve Bayes (Scratch)	Improvement
Accuracy	94.26%	**97.94%**	+3.68%
Precision	85.40%	**91.57%**	+6.17%
Recall	72.67%	**94.41%**	**+21.74%**
F1-Score	78.52%	**92.97%**	+14.45%

4.2. On Confusion Matrix

Metric	Decision Tree (Phase 1)	Naïve Bayes (Phase 2)	Definition
True Positives (TP)	117	**152**	Spam correctly identified.
True Negatives (TN)	934	**940**	Ham correctly identified.
False Positives (FP)	20	**14**	Ham incorrectly flagged as spam.
False Negatives (FN)	44	**9**	Spam missed (flagged as ham).

5. Conclusion

In phase 2, Naïve Bayes was implemented from Scratch. The Naïve Bayes model significantly outperformed the Decision Tree, particularly in Recall (increasing from 72.6% to 94.4%). In Confusion Matrix it can also be visualized. While a Decision Tree looks for specific "split words" (like "call" or "txt"), Naïve Bayes considers the collective probability of all words in the message, making it much more robust for detecting varied spam patterns